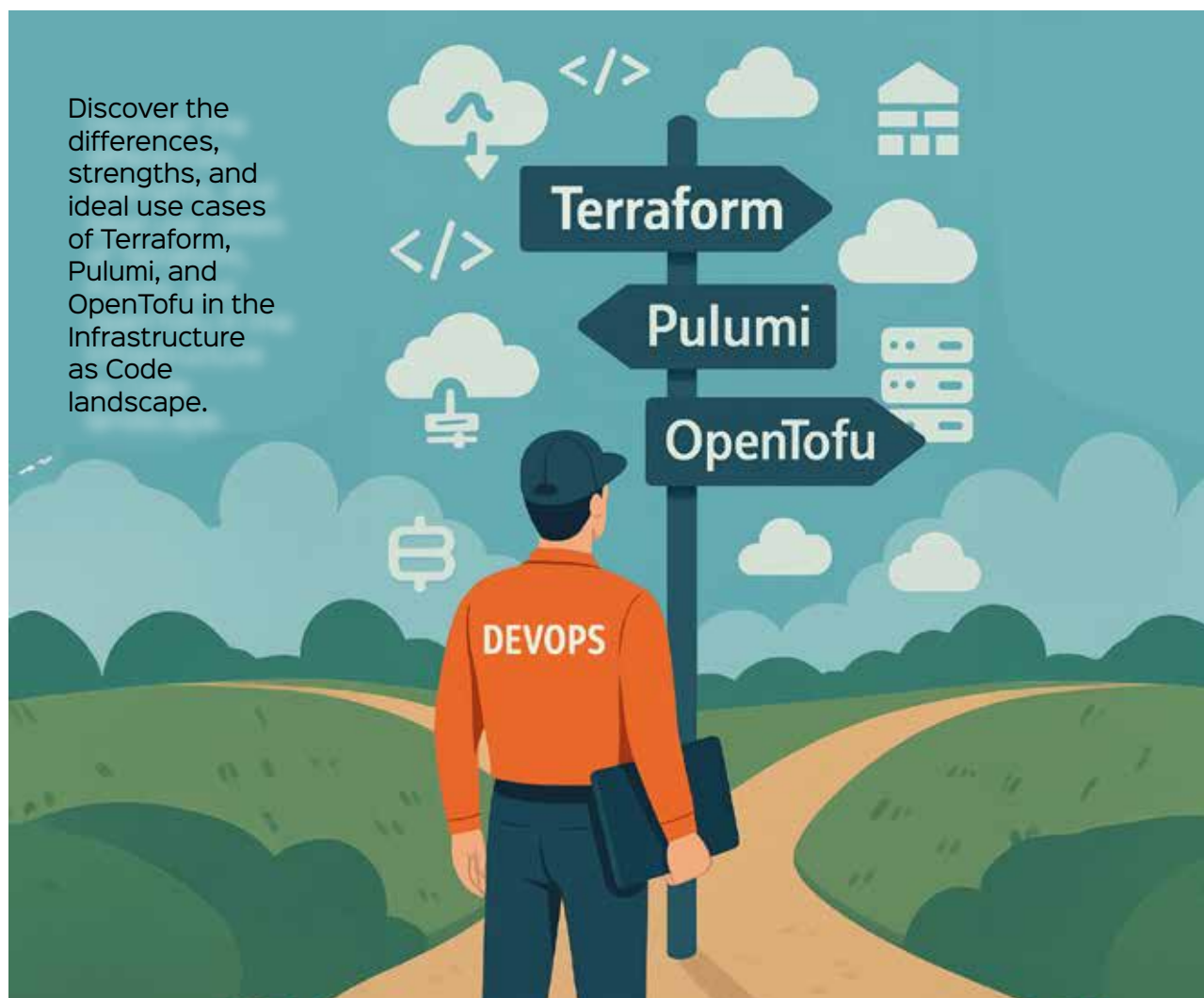


How to Choose Between Terraform, Pulumi, and OpenTofu



Discover the differences, strengths, and ideal use cases of Terraform, Pulumi, and OpenTofu in the Infrastructure as Code landscape.

Infrasturcture as Code (IaC) enables DevOps and cloud teams to manage infrastructure through code instead of manual processes. It ensures consistency, repeatability, and automation in infrastructure provisioning. In the IaC landscape, Terraform, Pulumi, and the open source OpenTofu stand out as the most talked-about tools.

Programming model and language support

Terraform uses a declarative approach through HCL (HashiCorp Configuration Language). It's purpose-built for defining infrastructure and is relatively easy to learn for infrastructure teams. However, it's not ideal for complex logic or abstractions.

Pulumi, on the other hand, uses

general-purpose programming languages such as Python, JavaScript, TypeScript, Go, and C#, combining both declarative and imperative models. This flexibility empowers developers to reuse existing libraries, implement logic, and write unit tests.

OpenTofu, a fork of Terraform, retains the HCL-based declarative style but focuses on staying fully open source